

STA510 - Mandatory Assignment 1

Tommaso Facchin

21 settembre 2026

Contents

1 Problem 1	1
1.1 a)	1

1 Problem 1

1.1 a)

For $f(x; a)$ to be a valid probability density function, it must be non-negative and integrate to one.

For $a \leq x \leq a + 20$, both factors are non-negative:

$$x - a \geq 0, \quad a + 20 - x \geq 0.$$

Since $c > 0$, it follows that

$$f(x; a) \geq 0.$$

To find the normalising constant, we require

$$\int_{-\infty}^{\infty} f(x; a) dx = 1.$$

Since the density is zero outside $[a, a + 20]$,

$$\int_a^{a+20} c(x - a)(a + 20 - x) dx = 1.$$

Using the substitution

$$u = x - a, \quad dx = du,$$

the limits become

$$x = a \Rightarrow u = 0, \quad x = a + 20 \Rightarrow u = 20.$$

Also,

$$a + 20 - x = a + 20 - (u + a) = 20 - u.$$

Therefore,

$$1 = c \int_0^{20} u(20 - u) du.$$

$$1 = c \int_0^{20} (20u - u^2) du.$$

$$1 = c \left[10u^2 - \frac{u^3}{3} \right]_0^{20}.$$

$$1 = c \left(10 \cdot 20^2 - \frac{20^3}{3} \right).$$

$$1 = c \left(4000 - \frac{8000}{3} \right) = c \frac{4000}{3}.$$

Hence,

$$\boxed{c = \frac{3}{4000}}.$$

Thus, the valid pdf is

$$f(x; a) = \begin{cases} \frac{3}{4000}(x - a)(a + 20 - x), & a \leq x \leq a + 20, \\ 0, & \text{otherwise.} \end{cases}$$